STAFF REVIEW · H446 032 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. Q becomes 1 at the rising edge because the flip-flop samples D. It then holds that value until the next triggering edge.

2. We group neighbouring 1s to simplify the Boolean expression. Variables that change within the group can be removed.

3. D is at the top. From top to bottom the stack is D, B, A. C was removed by the pop.

4. 13 = 00001101_2. Invert to 11110010_2, then add 1: 11110011_2. Prefixing the magnitude gives sign-and-magnitude (10001101_2), not two’s complement.

I Do — source worked steps

1. Create a console program — CODE:
```
using System;


Console.Write("Username: ");
string username = Console.ReadLine() ?? "";
```

2. Add the second input — CODE:
```
Console.Write("Demo password: ");
string password = Console.ReadLine() ?? "";
```

3. Plan, then add selection — CODE:
```
// Write an if statement:
// username must equal "student"
// AND password must equal "demo123"
// Print "Access granted" or "Access denied".
```

We Do — guided CHECK

1. Build the condition
CODE:
```
if (username == "student" && password == "demo123")
{
Console.WriteLine("Access granted");
}
else
{
Console.WriteLine("Access denied");
}
```

2. Explain your solution first
How could you allow at most three attempts?

Handle repeated attempts
CODE:
for (int attempt = 1; attempt <= 3; attempt++)
{
// Read inputs, check both values, break on success.
}

3. Explain your solution first
A student writes: 'Which structure is best?' Why is this too vague?
MCQ quality check
Specify the operation, such as key-based lookup, FIFO removal or index access, so there is one defensible correct answer.

You Do — source mark schemes

1. CODING 1 · SEQUENCE
Model: using System; read the username and demonstration password using Console.ReadLine() ?? "". The task is complete only when the program compiles and handles both inputs. Never use real passwords.

2. CODING 2 · SELECTION
if (username == "student" && password == "demo123") Console.WriteLine("Access granted"); else Console.WriteLine("Access denied");. AND is essential: one matching value alone is insufficient.

3. CODING 3 · ITERATION
for (int i = 0; i < 3; i++) { read both credentials; if (both match) { grant access; break; } if (i == 2) display lock message; }. Test successful attempt 1, success attempt 3 and three failures.

4. SYNOPTIC MCQ BANK
MCQ exemplars: 3D array arr[2,3,4] holds 24 entries; record groups named fields; a list allows a variable-length collection; a tuple groups a fixed set of possibly different types. Each authored MCQ must identify one correct response and justify distractors.

Exit Ticket — source answer

1. False. A login should check both the username and password before allowing access.